Q.1-) $(num \odot (\alpha \to \alpha)) \odot num$

$(\alpha \to num) \odot (\beta \to num)$

Q.2-) $(fun\ (x : \alpha) \to fun\ (y : \alpha \odot \beta) \to match\ y\ with\ \langle x \| y \rangle \to x)$ (w)

Q.3-)

$$\frac{\Gamma \vdash e_1 : \tau_1 \quad \Gamma \vdash e_2 : \tau_2}{\Gamma \vdash \langle e_1 \| e_2 \rangle : \tau_1 \odot \tau_2}$$

$$\frac{\Gamma \vdash e_1 : \tau_1 \odot \tau_2 \quad \Gamma, x : \tau_1, y : \tau_2 \vdash e_2 : \tau_3}{\Gamma \vdash match\ e_1\ with\ \langle x \| y \rangle \to e_2 : \tau_3}$$

Q.4-)

Histoire de capture :

$$\frac{y \in fv(v_2)}{match\ \langle v_1 \| v_2 \rangle\ with\ \langle x \| y \rangle \to e \longrightarrow e\{v_1/x\}\{v_2/y\}}$$

Q.5-)

$match\ \langle 2+2 \| (\lambda x.y)\,4 \rangle\ with\ \langle x \| y \rangle \to \lambda z.x+1.$

$\Rightarrow match\ \langle 4 \| (\lambda x.y)\,4 \rangle\ with\ \langle x \| y \rangle \to \lambda z.x+1.$

$\Rightarrow match\ \langle 4 \| y \rangle\ with\ \langle x \| y \rangle \to \lambda z.x+1 \Rightarrow \lambda z.4+1.$

Q.6-)

$$\frac{e_1 \Downarrow v_1}{\langle e_1 \| e_2 \rangle \Downarrow \langle v_1 \| v_2 \rangle}$$

$$\frac{e_1 \Downarrow \langle v_1 \| v_2' \rangle \quad e_2\{v_1/x\}\{v_2/x\} \Downarrow v_3}{match\ e_1\ with\ \langle x \| y \rangle \to e_2 \Downarrow v_3}$$

Q.7-) $\langle 1+1 \| 2+2 \rangle \longrightarrow \langle 1+1 \| 4 \rangle$

$\longrightarrow \langle 2 \| 2+2 \rangle$

Q.8-) oui (1 règle / construction) :

$$val\ \frac{}{v \Downarrow v} \qquad \frac{u \Downarrow v \quad t \vdash \lambda x.t' \quad t'\{v/x\} \Downarrow v'}{t\,u \Downarrow v'}$$

Q.9-)

Si $t \Downarrow v_1$ ou $t \Downarrow v_2$

- cas de base val : $t = v_1 \Downarrow v_1 \quad v_1 \Downarrow v_2 \quad v_1 = v_2$.
- par induction : App

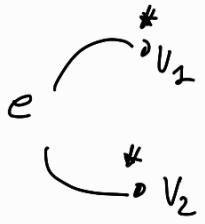
$$\frac{u_2 \Downarrow v \quad u_1 \Downarrow \lambda x.u_3 \quad u_3\{v/x\} \Downarrow v_1}{t = u_1\,u_2 \Downarrow v_1} \qquad v = v' \quad u_3 = u_3'$$

$$v_1 = v_2$$

$$\frac{u_2 \Downarrow v' \quad u_1 \Downarrow \lambda x.u_3' \quad u_3\{v'/x\} \Downarrow v_2}{t = u_1\,u_2 \Downarrow v_2}$$

Q.10-)

- e ⇓ v_1
- e ⇓ v_2



Il faudrait supposer v_1 et v_2 de même longueur.

- $$\frac{e_1 \to e_1'}{\langle e_1 \| e_2 \rangle \to \langle e_1' \| e_2 \rangle} \qquad \frac{e_2 \to e_2'}{\langle e_1 \| e_2 \rangle \to \langle e_1 \| e_2' \rangle}$$

- $$\frac{e_1 \overset{\|}{\to} e_1'}{\langle e_1 \| e_2 \rangle \overset{\|}{\to} \langle e_1' \| e_2' \rangle}$$

Par Induction

$e \overset{\|}{\to} e'$

- $\langle e_1 \| e_2 \rangle \to \langle e_1' \| e_2 \rangle \to \langle e_1' \| e_2' \rangle$

- $$\frac{e_2 \overset{\|}{\to} e_2'}{\langle v \| e_2 \rangle \to \langle v \| e_2' \rangle} \longrightarrow \frac{e_1 \overset{\|}{\to} e_1'}{\langle e_1 \| v \rangle \overset{\|}{\to} \langle e_1' \| v \rangle}$$